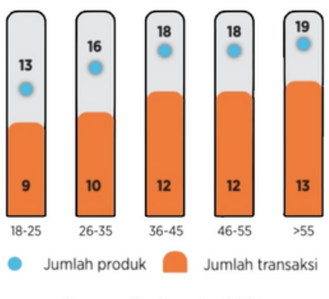
Prototype Development of a Food E-Commerce Application for the Elderly Using a User-Centered Design Approach

Jason Rivalino
13521008
13521008@std.stei.itb.ac.id

Adi Mulyanto
19631126 198803 1 002
adi@staff.stei.itb.ac.id

Background

Average number of transactions and product purchases per year by age group



Source: Beritasatu, 2021

The elderly dominate e-commerce transactions with a total of **13 transactions and 19 product purchases per year on average, the highest** compared to other age groups.

E-Commerce is important for the elderly to make the **transaction process easier** because their physical condition is **not as good** as that of younger people, and their daily needs **also differ.**

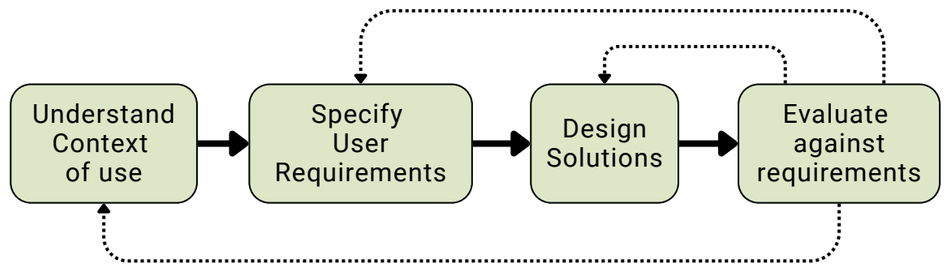
Healthy food is important to support elderly **physical function and mental health,** while also reducing the **risk of chronic diseases and disabilities.**

Existing platforms **only accommodate the needs of general users** and often **fail to consider elderly users,** who require **simpler ordering processes** and **interface designs with color schemes that do not hinder visibility.**

There is a need to **develop a food e-commerce application** to **help the elderly to order healthy food more easily** with **User-Centered Design approach.**

Methodology

Source: ISO 9241-210:2010



Data Gathering

Questionnaire
- Used to collect quantitative data from a larger sample size (Cresswell, 2014).
- Implemented by creating a Google Form, which was filled out by 71 respondents.
- Data transformation was applied to enhance the accuracy of problem identification.

Interview
- Used to collect in-depth and detailed information (Rachmawati, 2007).
- Involved preparing questions about the use of e-commerce for purchasing daily food, which were asked to 5 respondents during interviews.

Observation
Collect additional information by examining the current condition of existing applications.

Characteristics

- Users frequently use e-commerce platforms to shop or purchase products as part of their daily activities.
- Users want an e-commerce platform that offers convenience in the transaction process.
- Users want clear delivery of information regarding product listings or notification messages so that they are easy to understand.
- Users want an application design that is consistent and aligned with their functional needs.
- Users want flexibility in purchasing product needs, whether it is ready-to-eat meals or groceries that can be prepared at home.

Problems

- Difficulty in obtaining detailed information about products being sold, such as images, availability, safety labels, and the address of the business selling the products.
- Issues in the transaction process flow, such as difficulties in customizing orders and completing the payment process.
- Application design issues, particularly in menu layout and color schemes that are unsuitable for elderly users.
- Difficulty in text readability for elderly users due to font size, typeface, and color contrast.
- Unclear delivery information, including driver location, estimated arrival time, and difficulty contacting the driver.
- Security concerns, as the application is not yet fully trustworthy.

User Requirements

- Users gain personal security and trust in using the application in an easy way.
- Users can search for businesses and order desired products more easily.
- Users can carry out the product ordering transaction process more easily.
- Users can access information about the delivery of orders they have previously made.
- Users can receive clear notification information when performing actions, with a display that adapts to their needs.
- Users can view information related to their previous purchase history.
- Users can provide ratings and comments to the driver or business they ordered from previously.

Apps Activity

- User account management
- Business exploration and selection
- Product selection and customization
- Order confirmation
- Checking order information
- Canceling orders
- Contacting the driver
- Business rating and review

Usability Goals

Effective to use | Efficient to use | Safe to use | Easy to learn | Enjoyable to use

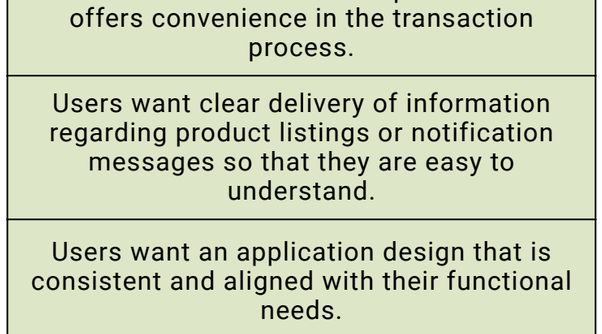
User Experience Goals

Helpful | Engaging

Low-Fidelity Prototype

Provide an overview of interaction design and a detailed application flow.

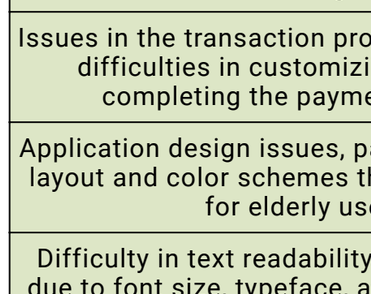
Tech Stack: Figma | Total: 30+1 prototypes



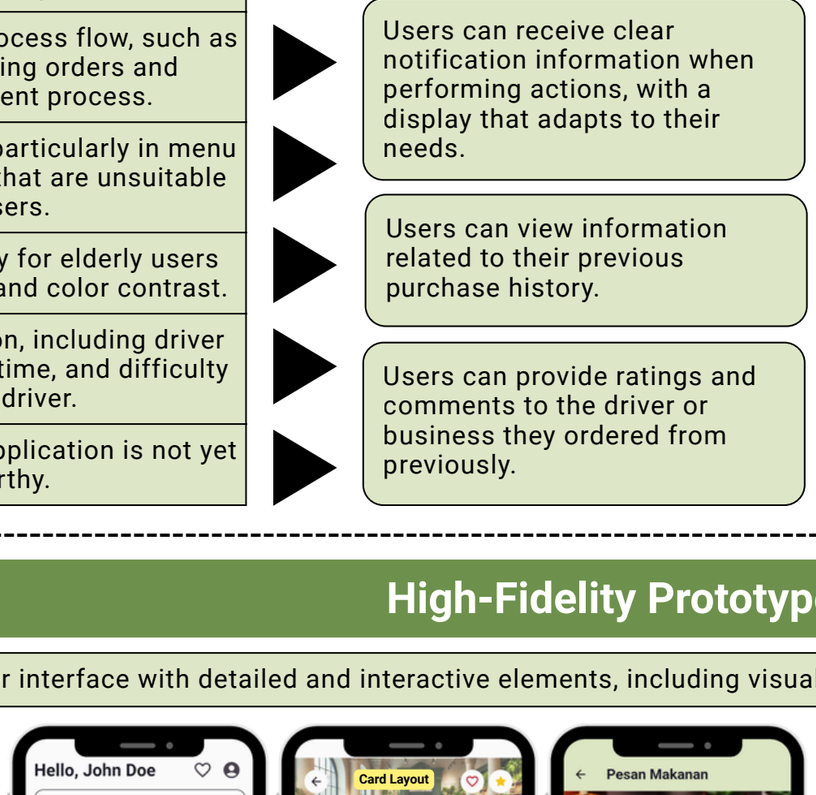
High-Fidelity Prototype

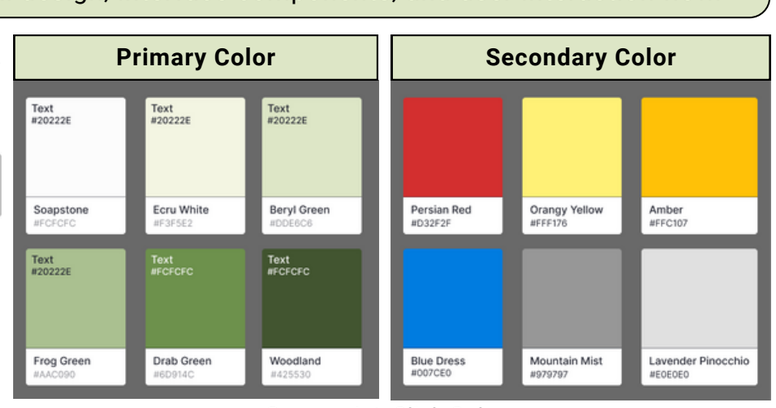
Represent a realistic user interface with detailed and interactive elements, including visual design, interface components, and user interaction flow.

Tech Stack: Flutter & Dart



Nutrisenior Logo
Source: Brandcrowd





Source: ArtyClick Colors

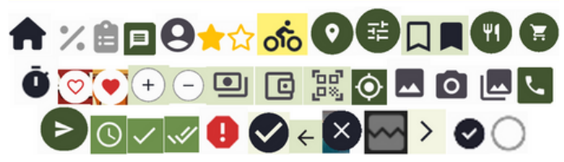
Font Size

Main Title (40dp - 32px)
Heading (28dp - 21px)
Heading (24dp - 18px)
Heading (22dp - 17px)
Heading (20dp - 15px)
Heading (18dp - 14px)
Paragraph (16dp - 12px)
Paragraph (14dp - 11px)
Paragraph (13dp - 10px)

Bold Weight

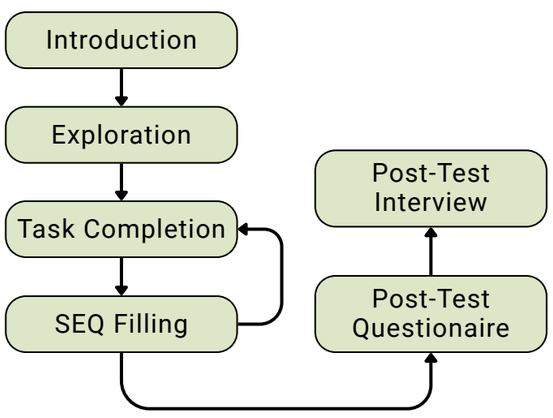
Paragraph (Bold) w700 | Paragraph (Medium) w500 | Paragraph (Regular) w400

Iconography



Evaluation

Obtain user feedback on the quality, comfort, and relevance of the developed interface design to user needs



Goals	Metrics	Iteration I	Iteration II	Improvement
Effective to use	Task Completion Rate	99,375%	100%	0,629%
Efficient to use	Time on Task	993,32s	822,55s	17,19%
Safe to use	Number of Errors	1	0	100%
Enjoyable to use	Customer Satisfaction Score	100%	100%	0%
Easy to learn	Single Ease Questions	6,875	7	1,82%
Helpful	IMI (Value/Usefulness)	6,86	7	2,08%
Engaging	System Usability Scale	94,5	100	5,82%

Technical Documentation:
https://drive.google.com/file/d/11QIss4Sy9Rr5Bxy_UVJZ-1W2w4VSf0EU

Figma Prototype:
https://www.figma.com/design/gJuODaLsdT8PQrgCz9ttUP/NutriSenior-App-for-TA?node-id=0-1&t=xJcGBQeaZ231TDM5-1

Frontend Github Repository
https://github.com/jasonrivalino/ta_nutrisenior_app